UFCFXK-30-3: Digital Systems Project

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Project Title: **Predictive Spice-Stock Preservation System for Warehouse Environments**

Watch the demo video!

Abstract:

This project develops a **predictive spice-stock preservation system** for warehouses using distributed temperature/relative-humidity sensor pods and a **gateway analytics pipeline**. It detects short-horizon environmental risk by collecting multi-point data, running preprocessing plus forecasting/anomaly detection, and presenting actionable dashboards/alerts. Temperature and RH are used because they dominate real-time food monitoring and are strongly linked to moisture-driven quality deterioration in stored products.

Aim: Build an energy-aware, multi-point warehouse monitoring system that predicts adverse T/RH conditions and communicates risk clearly.

Objectives:

- Build a reliable T/RH sensor pod deployable across multiple zones.
- Implement gateway ingestion, preprocessing, and timestamped CSV logging.
- Implement forecasting/anomaly detection and evaluate against simple baselines.
- Deliver a lightweight dashboard showing readings, trends, and alerts.
- Document RF/missing-sample issues and deployment constraints.

Research:

Real-time monitoring is widely used in storage/transport to **reduce food loss and waste**, with **temperature and humidity** as the most common variables. (da Costa et al., 2022) Warehouse **IoT systems** demonstrate environmental **monitoring/control**, and recent work adds intelligent decision logic (e.g., fuzzy/learning) for improved responsiveness. (Parkhi et al., 2022; Zhang et al., 2025; Gowrishankar et al., 2023) For spices, **moisture sorption** and **water-activity–temperature** relationships can cause physical instability (e.g., **caking**), motivating early detection. (Voelker et al., 2020) Warehouse **RF** conditions can **reduce BLE** reliability (metal/multipath) and 2.4 GHz coexistence can add interference; scaling many peripherals also raises gateway scheduling/fairness issues. (Ramires et al., 2024; Eltholth, 2023; Kim & Paek, 2025) Over time, concept drift can degrade static models, supporting logging and periodic re-validation. (Carnier et al., 2025)

Functional Requirements:

Measure and timestamp T/RH at configurable intervals.
Transmit pod readings to a gateway reliably.
Log data locally (CSV) for evaluation/diagnosis.
Run forecasting/anomaly flags and generate alerts.
Provide a dashboard for status, trends, and alerts.

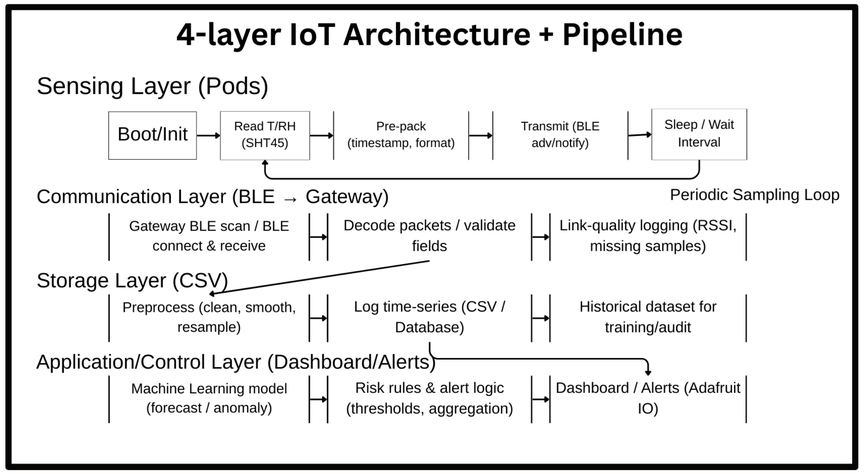
Non-functional Requirements:

Battery-aware pod operation (low duty cycle).
Robust under intermittent connectivity (gateway keeps logging).
Scales to multiple pods with predictable performance.
Low-noise alerting (avoid alert storms).

Design, Implementation & Testing:

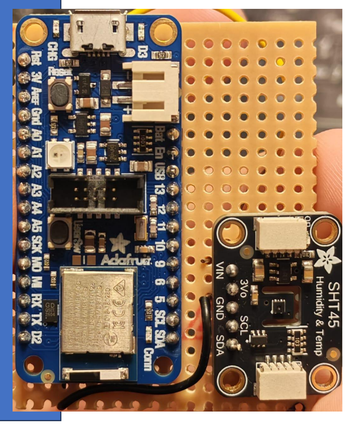
The system uses a layered pipeline: pods measure T/RH and forward readings to a gateway; the gateway validates and preprocesses data, runs forecasting/anomaly detection, and logs time-series for evaluation; the application layer visualises trends and alerts. The current artefact is a physical pod built from a BLE-capable board and a high-accuracy T/RH sensor. Next steps focus on stable gateway transfer, structured logging, and baseline analytics before extending models and alert logic.

State *Diagram*



Implementation Artifacts:

- **Board:** Adafruit Feather nRF52840 Express (BLE, ARM Cortex-M4F, USB).
- **Sensor:** Adafruit Sensirion SHT45 (high-accuracy digital T/RH, I^2C).
- **Relevance:** Supports low-power, reliable warehouse sensing pods.



Testing includes: (a) sensor stability/sanity checks, (b) comms reliability tests (distance, obstruction, missing samples), and (c) forecasting evaluation using standard error metrics and baseline comparisons. Where possible, multi-pod trials will assess scaling and confirm that logging supports diagnosis and reproducible evaluation.

Planning and Management:

Milestones: (1) hardware validation, (2) gateway ingestion + CSV logging, (3) baseline dashboard integration, (4) forecasting/anomaly evaluation, and (5) packaging/documentation/demo. Main risks are wireless reliability and time; mitigation is to deliver a minimal end-to-end pipeline first, then iterate once stability is proven.

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